

PHASE

PORJECT DEVELOPMENT PHASE

Date	28 September 2026
Team ID	SWTID-2026-5127
Project name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum marks	4

5. Project Development Phase

5.1 Purpose of this phase

This is where the design is implemented in ServiceNow. The phase covers preparation of data, building the catalog item, creating the flow and subflows, configuring notifications, and assembling reports. Development follows the plan sprint by sprint and is reviewed continuously.

5.2 Step 1: Prepare the environment and test data

1. Log in to the ServiceNow instance as an administrator.
2. Create a **dedicated application scope** (or use an update set) so all changes are grouped and portable.
3. Create groups: **Procurement Team** and **IT Fulfilment Team**, and add members.
4. Create sample users: a requester, a manager (set as the requester's manager), a user without a manager, and group members.
5. Assign the required roles (for example, approver and catalog task roles).
6. Confirm outbound email is configured for notification testing.

5.3 Step 2: Create the catalog item

1. Navigate to **Service Catalog > Catalog Definition > Maintain Items** and click **New**.
2. Enter the name "Standard Laptop Request", assign it to the **Hardware** category and add a short description and icon.
3. Set availability using user criteria so all employees can see it.
4. Save the item, then add variables in the **Variables** related list:

Order	Variable name	Type	Mandatory
100	requested_for	Reference (sys_user)	Yes
200	laptop_model	Select Box	Yes
300	justification	Multi Line Text	Yes
400	delivery_location	Reference (cmn_location)	Yes
500	needed_by	Date	Yes
600	cost_center	Reference / String	Yes
700	accessories	Checkbox	No

5. Add the **question choices** for the laptop model select box, using the approved standard models.
6. Add a **Catalog Client Script** (onChange on `requested_for`) to populate the cost center, and an **onSubmit** script to prevent past dates for `needed_by`.
7. Publish the item and confirm that it appears in the portal.

5.4 Step 3: Build the main flow in Flow Designer

1. Open **Flow Designer** and click **New > Flow**.
2. Name it **Standard Laptop Order Automation**, set the application, and choose to run as the appropriate user.
3. **Trigger:** select **Service Catalog** and choose the "Standard Laptop Request" item. The trigger provides the request item record and its variables as data pills.
4. **Action – Look Up Record:** find the user record from the `requested_for` variable so the manager is available.

5. **Flow logic – If:** check whether the manager is empty. In the *true* branch, set an approver variable to the fallback approver; in the *false* branch, use the manager.
6. **Action – Update Record:** set the RITM stage/state to *Awaiting Approval*.
7. **Action – Ask for Approval:** approver from the previous step; set the rule to *Approve* when the approver approves, *Reject* when they reject; include a clear message with model, justification and needed-by date via data pills.
8. **Flow logic – If:** branch on the approval state.

On approval:

- **Create Catalog Task** assigned to the Procurement Team, with a short description such as "*Order [laptop model] for [requested for]*" and a description containing delivery location and cost center.
- **Update Record:** RITM state to *Work in Progress*.
- **Send Notification/Email** to the requester confirming approval.
- **Wait for Condition:** wait until the procurement task state is *Closed Complete*.
- **Create Catalog Task** for IT Fulfilment (imaging, software installation, handover).
- **Wait for Condition:** wait until the IT task is closed.
- **Update Record:** RITM state to *Closed Complete*.
- **Send Email:** completion message to requester.

On rejection:

- **Update Record:** RITM state to *Closed Incomplete* and write the approver's comment.
 - **Send Email:** rejection message including the reason.
9. Save and **Test** the flow using the built-in test option before activating, then **Activate**.

5.5 Step 4: Add reminders and escalation

1. After the Ask for Approval action is started, add a parallel path (or use a timed check) with a **Wait for a duration** action, for example two business days.
2. After the wait, check whether the approval is still *Requested*.
3. If so, **Send Email** as a reminder to the manager.
4. Optionally add a second, longer wait, followed by escalation to the manager's manager or the fallback approver.

5.6 Step 5: Create reusable subflows

- **Subflow: Manager Approval with Fallback** – inputs: user, request record; outputs: approval decision and comments.
- **Subflow: Create Fulfilment Task** – inputs: assignment group, short description, description, due date; output: the created task.

The main flow then calls these subflows, reducing duplication and making the flow easier to read and to reuse for future catalog items.

5.7 Step 6: Configure notifications

1. Go to **System Notification > Email > Notifications** or use the Flow Designer *Send Email* action for flow-specific messages.
2. Write clear subjects such as "*RITM0010023 – Your laptop request has been approved.*"
3. Use dynamic content (requester name, request number, model, approver comments).
4. Include a link to the request in the portal.
5. Confirm that emails appear in the **Email Logs** during testing.

5.8 Step 7: Error handling

- Use the flow's **error handling** to catch failures, notify the admin group and mark the RITM for manual attention.
- Verify data pills are not empty before use in critical actions (for example, group or manager).
- Keep the execution details available for troubleshooting.

5.9 Step 8: Reports and dashboard

Create reports on the `sc_req_item` table filtered on the catalog item:

- Count of requests per month.
- Average time between creation and closure.
- Rejection rate.
- Requests by laptop model.

Add them to a dashboard for IT management.

5.10 Step 9: Package the work

All configuration is collected in an **update set** (or scoped application) so that it can be exported, reviewed and promoted to other instances. The update set is named clearly, for example *"Laptop Order Automation v1.0"*.

5.11 Development standards followed

- Descriptive, consistent names for flows, actions and variables.
- Comments on complex logic.
- Minimal scripting; out-of-the-box actions preferred.
- No hard-coded user names or sys_ids where a data pill or property can be used.
- Regular saving and versioning of the flow.

5.12 Sample configuration summary

Component	Configured value
Catalog item	Standard Laptop Request
Trigger	Service Catalog
Approver	Requested-for user's manager (fallback: designated approver)
Procurement group	Procurement Team
Fulfilment group	IT Fulfilment Team
Notifications	Submission, approval, rejection, reminder, completion
Deployment vehicle	Update set

5.13 Challenges faced and how they were handled

Challenge	Resolution
Manager not populated for some users	Added fallback approver logic and a data-quality check
Data pills not available after changing the trigger	Re-selected the trigger and refreshed pills
Tasks waiting indefinitely	Added a reminder and defined ownership on the task group
Notification content hard to read	Simplified wording and added dynamic request details

5.14 Outcome of the phase

A functioning solution: catalog item published, flow activated, tasks and notifications configured, reports created and all changes captured in an update set. The system is ready for structured testing.